13 November 2026

Dr Matteo Giacomo Prina

Academic Editor, PLOS Climate

**Re: Revised manuscript PCLM-D-26-00445**, “Kaya-guided hybrid deep learning for leave-country-out estimation of national $CO_2$ emissions: a 112-country benchmark with country-clustered inference, conformal uncertainty, and explainable AI”

Dear Dr Prina,

Thank you for the opportunity to revise our manuscript. We have addressed every point raised by the two reviewers and every journal requirement. The accompanying Response to Reviewers answers each comment individually, including the 36 comments in Reviewer 2’s annotated file. The manuscript has been restructured substantially. The Introduction now ends with four research questions; a workflow figure and a study-design section introduce the experiments before they are used; Methods, Results and Discussion are strictly separated; and the main text contains 7 instead of 34 numbered equations and 8 instead of 12 figures, with standard material and the supporting temporal experiment moved to Supporting Information. In response to the reviewers we added sensitivity analyses of the missing-data treatment and of the atypical economies, two further measures of covariate importance, a like-for-like test of the value of the low-carbon share, and an analysis that explains the coverage of the prediction intervals. The central finding, that neither deep nor hybrid models were shown to outperform ridge regression, is now stated plainly throughout, and the Discussion explains why and gives a roadmap of recommendations.

While revising we re-derived the panel from the public source files and re-checked every statement. This led us to correct several errors and imprecisions of the original submission,

which we list openly at the beginning of the Response to Reviewers. None of them changes the conclusions.

As requested, we provide the following statements for the submission form.

**Competing interests.** All authors have confirmed. I have read the journal's policy and the authors of this manuscript have the following competing interests: S.C.A. is a director of Premium Intelligence Networks and Systems Co. Ltd., Accra, Ghana, a private company. The company had no role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript, and has no commercial interest in the results. This does not alter our adherence to PLOS Climate policies on sharing data and materials. The other authors have declared that no competing interests exist.

**Financial disclosure.** All authors have confirmed. The authors received no specific funding for this work.

**Data availability.** All data underlying the findings are publicly available. The harmonized panel, the out-of-fold estimates of every model, the values behind every table and figure and all code are available from Zenodo at insert DOI. The source data are the Our World in Data $CO_2$ and energy data sets (https://github.com/owid/co2-data and https://github.com/owid/energy-data; repository versions of 2 June 2026 and 27 April 2026), which redistribute the Global Carbon Budget 2025, the Energy Institute Statistical Review of World Energy 2025, the International Energy Data of the U.S. Energy Information Administration and the Maddison Project Database 2023.

**Use of AI tools.** The statement required by the journal's policy is given in the Acknowledgments of the manuscript and is reproduced in the Response to Reviewers.

The revised submission consists of the Response to Reviewers, the revised manuscript with tracked changes, the clean manuscript, eight figure files in TIFF format, and ten Supporting Information files (S1 Text, S2 Text, S1 Table, S2 Table and S1–S6 Fig). All authors have approved the revised manuscript.

Yours sincerely,

Samiuel C. Atuahene (corresponding author), on behalf of all authors

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